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IRIT
Institut de Recherche en Informatique de Toulouse
CNRS - INP - UT3 - UT1 - UT2J

UNIVERSITÉ TOULOUSE CAPITOLE
FACULTÉ D'INFORMATIQUE

Black-box Active Learning for Discovering New Explanations

MAC

A practitioner's perspective

MIMICO

Derivative-Free Optimization Symposium

June 25th, 2026

Paul SAVES, Nicolas VERSTAEVEL, Benoît GAUDOU

Context

SMAC Team: 30+ Members

- *Multi-agent systems & simulation*
- *Explainable AI for complex systems*
- *Socio-environmental digital twins*

Institut de Recherche en Informatique de Toulouse
CNRS - Toulouse INP - UT3 - UT Capitole - UT2

600 lab members
Yearly budget 5.1M€ (2023)

Membre de l'Université Toulouse Capitole

ANR-JCJC MIMICO: MIMIcking the COmplexity of agent-based models

Machine-Learning

Agent-based simulation

Team

Nicolas Verstaevel - Principal Investigator

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Extended team

Manon Prédhumeau

Benoit Gaudou

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MIMICO

Membre de l'Université Toulouse Capitole

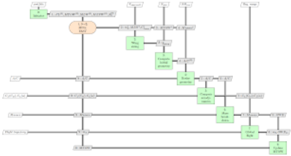
Created in 1229
3 campus
750 Teaching staff
740 administrative staff
21000 students
151 national diplomas

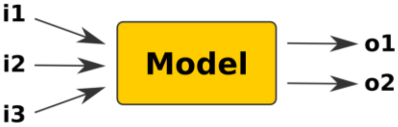
https://www.irit.fr/mimico/ & Nicolas.Verstaevel@irit.fr

Complex Systems Simulations

- **Design-based simulations**
  - Useful to study complex multidisciplinary engineering challenges (aerospace design, structural mechanics, fluid dynamics, thermal management, energy efficiency, advanced manufacturing…)
  - Simulation models can be abstracted as a program (or a set of coupled solvers) that transforms a set of design **inputs** into a set of performance **outputs**.
  - ➢ **Sensitivity Analysis/Exploration/Optimization**: goal-oriented and user-defined automated exploration of a model’s outcomes to study a given phenomenon with trade-offs between competing disciplines.

https://fast-oad.readthedocs.io/





Plat, G., Saves, P., Bartoli, N., Morlier, J., **Towards frugal Bayesian optimization for environmental impact minimization of computer experiments**, MASCOT-NUM, 2026.

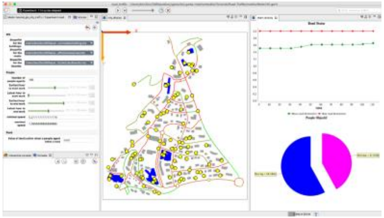
Socio-Technical Complex Systems Simulations

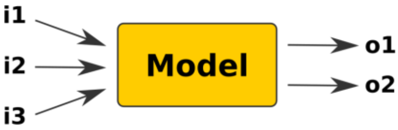
Agent-based simulations

- Useful to study **complex socio-environmental** challenges (climate change, water scarcity, urbanization, pollution, energy transition, sustainability…)
- Simulation models can be abstracted as a program that transforms a set of **inputs** into a set of **outputs**.

➢ **Calibration/Exploration/Optimization**: goal-oriented and user-defined automated exploration of a model's outcomes to study a given emerging and decentralized phenomenon

https://gama-platform.org/





Parsons, S.D., Gmytrasiewicz, P. and Wooldridge, M.J., **Game theory and decision theory in agent-based systems,** Springer Science & Business Media, 2002.

Interactive Platform & Decision-making

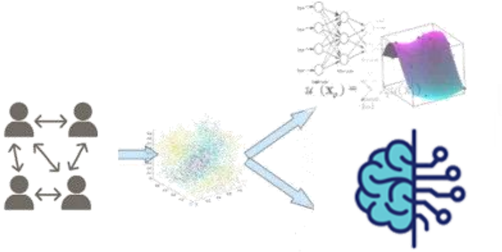
- Frugal AI: Overcoming the computational barrier to participatory systems through spatiotemporal and multi-fidelity surrogate models.

- Explainable AI: Ensuring algorithmic transparency to identify the system’s tipping points and build genuine public trust in the models.

Limits of Stochastic Complex Systems Simulations

- **High computation costs:** large numbers of agents inter. **over time and space**. *High-dimensional mapping via LHS is intractable.*
- **Opacity of outcomes:** intricate interactions among agents often lead to results that are **hard to trace back** to specific inputs or rules. *Not easily decomposable.*
- **Black-Box:** No gradients to guide the search towards interesting behaviors.
- **Needle in a Haystack:** Traditional Active Learning minimizes *prediction uncertainty*, wasting simulations in noisy but homogeneous regions while missing critical tipping points.

Explainable AI for Surrogate models



- **Mimic** a complex system, black box ⇒ ease calibration, scenarios exploration, and optimization
- Types of models used ⇒ Models are **complementary**: **complexity** or **explicability** based on **machine learning**

Forrester, A., Sobester, A., Keane, A., **Engineering design via surrogate modelling: A practical guide,** John Wiley & Sons, 2008.

Redefining Tipping Points in Complex Systems

From Dynamical Systems to Active Learning:

- **The Classical View:** In dynamical systems out-of-equilibria, a tipping point is a mathematical discontinuity (e.g., bifurcations, systems moving out-of-equilibria) where the state: $\lim_{\Delta_x \to 0} \frac{\Delta_y}{\Delta_x} = \infty$.
- **The Intractability:** Proving a true analytical discontinuity is impossible from finite samples in a stochastic, derivative-free, black-box ABM.
- **The Lipschitz Relaxation:** We mathematically relax this constraint. Instead of seeking an infinite gradient, we define a generalized tipping point as a localized region (a ball B) exhibiting an exceptionally high Lipschitz constant (extreme finite-difference variations).

Although predicting such critical tipping points is extremely difficult, work in different scientific fields is now suggesting the existence of generic early-warning signals that may indicate for a wide class of systems if a critical threshold is approaching.

Scheffer M, Bascompte J, Brock WA, Brovkin V, Carpenter S., Dakos V, Held H, Van Nes E., Rietkerk M, Sugihara G. **Early-warning signals for critical transitions.** Nature. 2009.
Cavagna, A., Cimarelli, A., Giardina, I., Parisi, G., Santagati, R., Stefanini, F. and Tavarone, R., **From empirical data to inter-individual interactions: unveiling the rules of collective animal behavior.** Mathematical Models and Methods in Applied Sciences, 2010.

Optimization goal: A Bi-Objective Problem

Our exploration strategy can be mathematically cast as a bi-objective Derivative-Free Optimization (DFO) problem over the sampling density $\rho(x)$:

- **Objective 1 (Global Discrepancy)** $\min_{x \in \chi} \rho(x|X)$

  We seek to maximize space-filling discrepancy to avoid pathological density peaks in the input space (preventing model collapse in homogeneous regions).

- **Objective 2 (Targeted Tipping Point Exploitation):**

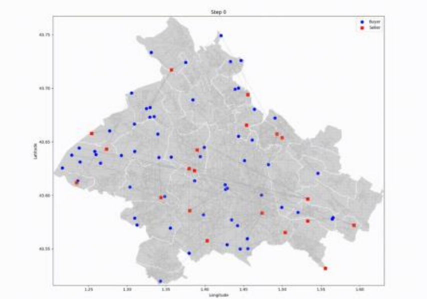
$$\max_{x \in \chi} \rho(\, x \mid x \in \mathcal{B}(x_{tp}, r), X)$$

  Simultaneously, we must locally maximize sampling density around the tipping points.

The Challenge: Solving this directly is intractable. $\mathcal{B}(x_{tp}, r)$ and $\rho \mid x \in \chi$ are to be discovered.

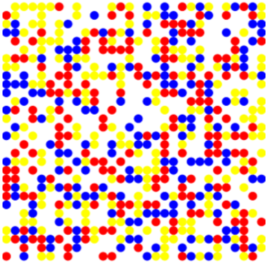
From Agent based simulations to emergent behaviors

- Industrial Symbiosis model - MASTIO
  - Initial random placement, RL algo., decreasing temp.



- 5 cont.: product scarcity, firm density, transport. cost, disposal cost, cluster spread

- Schelling segregation model
  - Initial random placement, random move if unsatisfied.



- 3 integers: grid size, number of socio-types, agent perception distance & 2 cont.: intolerance of unhabitant and population density.

https://mobiliscope.cnrs.fr/en/geoviz/montreal

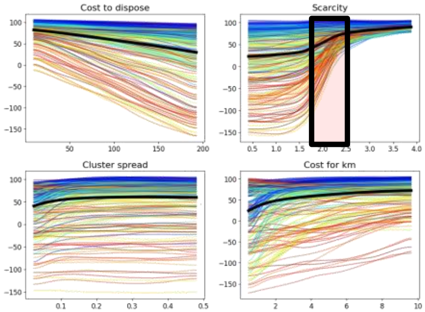
Schelling, T., **Dynamic models of segregation.** Journal of mathematical sociology, 1971.

Mastio, M., Saves, P., Gaudou, B., Verstaevel, N., **Adaptive Agents in Spatial Double-Auction Markets: Modeling the Emergence of Industrial Symbiosis,** AAMAS 2026.

Motivation: Tipping Points in Dual Spaces

Parameter Space (MASTIO ICE/PDP)

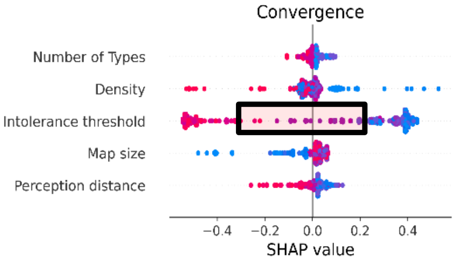
Regression



- Tipping point: Prediction collapses/diverges at specific parameter thresholds.
- Transient phases may exhibit scale-free (power-law) dynamics.

Explanatory Space (Schelling SHAP)

Classification



- Internal logic shift: Appearance of isolated "islands" where features suddenly dominate convergence.

RQ: Can hunting for explanatory novelty help find physical tipping points?

Saves, P., Gaudou, B., Verstaevel, N., **Modèles de Substitution pour les Modèles à base d'Agents: Enjeux, Méthodes et Applications**, JFSMA 2025.

Mastio, M., Saves, P., Gaudou, B., Verstaevel, N., **Adaptive Agents in Spatial Double-Auction Markets: Modeling the Emergence of Industrial Symbiosis,** AAMAS 2026.

Pipeline & Innovations

Surrogate-based Active Learning

Local explainability

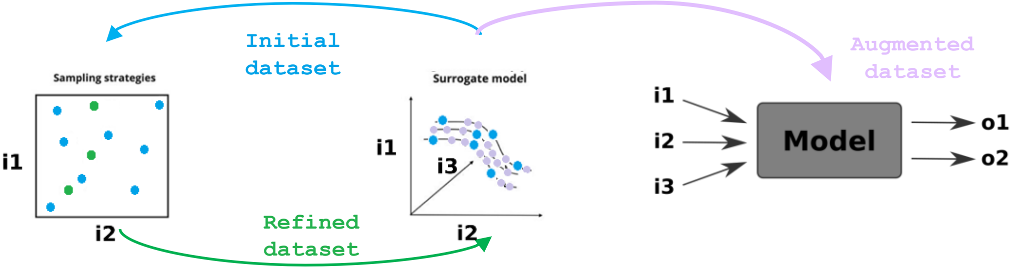
Q2: What does SM tell us about the parameter space?

`Exploration + Calibration`

Global explainability

Q3: How does the SM inform us about the model dynamics?

`Analysis + XAI`



Q1: How to efficiently train SM from a few samples?

`Initial + Refined datasets`

Pre-Processing: Handling ABM Stochasticity

Removing Aleatoric Uncertainty via Statistical Power Analysis.

To construct a reliable deterministic surrogate model for Active Learning, we "remove" noise by estimating the true mean response $\mathrm{y}(\mathrm{x}) \cong \hat{y}$.

- **The Challenge:** How many replications R are required to confidently estimate the mean response without wasting computational budget?
- **Our Solution:** We dynamically execute R replications until the variance of the mean stabilizes.
- **The Stopping Criterion:** This stabilization is verified online using an **F-test (Fisher)** over the running variance, ensuring sufficient statistical power and minimizing Type-II false negatives.

Is $\hat{y}(x) | \{y_1(x), | \, y_r(x)\} = \hat{y}(x) | \{y_1(x), | \, y_{r-1}(x)\}$?

Secchi, D., & Seri, R., **Controlling for false negatives in agent-based models: a review of power analysis in organizational research**. Computational and Mathematical Organization Theory, 2017.

Lorscheid, I., Heine, B. O., & Meyer, M**., Opening the 'black box' of simulations: increased transparency and effective communication through the systematic design of experiments**, Computational and Mathematical Organization Theory, 2012.

Constructing the Dual Explanatory Space

- **The Value Function $v(S)$:** For any feature coalition $S \subseteq M$, the value function is the conditional expectation:

$$v(S) = \mathbb{E}\left(f(X) \mid X_S = x_S\right)$$

- **The Local Payout:** The SHAP values $\phi_i(x)$ distribute the exact difference between the local prediction and the global average:

$$\sum_{i=1}^{M} \phi_{i(x)} = f(x) - \mathbb{E}\left(f(X)\right)$$

- **The Dual Mapping:** Each physical point $x \in \mathbb{R}^m$ is mapped to its local explanatory profile $S(x) \in \mathbb{R}^m$:

$$S(x) = \phi_1(x), \ldots, \phi_M(x)$$

Lundberg, S. M., Lee, S.-I., **A Unified Approach to Interpreting Model Predictions**, NeurIPS, 2017.
Robani, M., Palar, P., Zuhal, L., Saves, P., Morlier, J. , **SMT-EX: An explainable surrogate modeling toolbox for mixed-variables design exploration.** AIAA SciTech, 2025.

The Core Concept: Explanatory Diversity

Instead of looking only at the prediction *y*, we extract feature contribution profiles using post-hoc XAI.

SHAP values attribute, for a given prediction, how much each feature contributes to the deviation between that prediction and the model's average prediction, by fairly distributing that difference across features.

$$\phi_{i,j} = \sum_{U \subseteq [1:m] \setminus \{i,j\}} \frac{|U|!\,(m-|U|-2)!}{(m-1)!} \left(v(U \cup \{i,j\}) - v(U \cup \{i\}) - v(U \cup \{j\}) + v(U) \right),$$

- **Goal:** Explicitly sample regions where the explanation of the model fundamentally changes.

Shapley, L. S., **A value for n-person games,** Contributions to the Theory of Games, 1953.

Derivative-Free Min-Max Cosine Proxy

From Slow KDE to Fast Cosine Proxy:

- My initial idea relied on high-dimensional Kernel Density Estimation (KDE) on SHAP values:

$$\rho\big(S(x)\big) = \frac{1}{N}\sum_{i=1}^{N} K_h\big(S(x) - S(x_i)\big)$$

KDE suffers from the curse of dimensionality, decouples parameters, creates normalization issues, and is computationally prohibitive.

The Solution: We implemented a **Fast Density Proxy via Cosine Distance.** High cosine similarity (CS) to the nearest neighbor translates directly to high explanatory novelty.

The direction of the XAI change, even when compared to the most similar explanation!

→ ***Max-****CS to the Nearest Neighbor (1-NN) in SHAP space ~ "Max-Surprising"*

Palar, P.S, Saves, P., Robani, M.D, Garouani, M., Verstaevel, N., Aligon, J., Shimoyama, K., Gaudou, B., Morlier, J., **Interpretable and Explainable Surrogate Modeling for Simulations: A State-of-the-Art Survey and Perspectives on Explainable AI for Decision-Making**, Archives on Computational Methods in Engineering, 2026.

Dual-Surrogates & SHAP Data Augmentation

SMT-optim

Bypassing the ABM Computational Bottleneck:

- **Primary GP (Physical Space):** Trained directly on the true ABS outputs. It acts as an ultra-fast proxy for the simulation (*Dataset* $\mathcal{D}$).
- **Data Augmentation Trick:** ExactSHAP computations are exponentially expensive $\mathcal{O}(2^M)$. We use the Primary GP to generate thousands of cheap SHAP profiles (*Low-Fidelity XAI data*).
- **The Result:** A massive, augmented dataset combining a few exact ABM profiles (*High-Fidelity XAI data*) and thousands of GP-derived profiles, providing dense explanatory context evaluated to give *Low-Fidelity and High-Fidelity XAI Novelty*.
- **Secondary Multi-fi GP (XAI Space):** Trained on the 2-fidelities *XAI Novelty* sets, surrogating $\rho_{SHAP-CS}$.

Cordelier O, Diouane Y, Bartoli N, Laurendeau E. **Multi-fidelity constrained Bayesian optimization with application to aircraft wing design.** AIAA Aviation 2025.

Stepwise Uncertainty Reduction

From Uninformed Variance Reduction to IMSE:

- My initial idea relied on Variance of the GP reduction: $\min_{\Omega} Var_{GP}(x)$. Greedy variance reduction suffers from the curse of dimensionality, greedily targeting model boundaries but ignoring the global impact of adding a new sample.

The Solution: Integrated Mean Squared Error (IMSE). Instead of exploring blindly, we compute the sum of the expected global variance reduction. $\Delta IMSE(x) = \int_{\chi} Var_{GP}(u) \mid GP_{\mathcal{D}\cup(x,\mu(x))} - \int_{\chi} Var_{GP}(u) \mid GP_{\mathcal{D}}$

The Kriging Believer: To evaluate a candidate x, we update the GP with its own mean prediction at x. We then measure how much this hypothetical update collapses the uncertainty across the entire physical space.

→ ***Max***-*ΔIMSE* for robust Stepwise Uncertainty Reduction (SUR) ~ *"Max-Uncertainty Reducing"*

Picheny, V., **A stepwise uncertainty reduction approach to constrained global optimization**, Artificial Intelligence and Statistics, 2014.

Mixed-integer bi-objective DFO

Framing Dynamic-US as a Bi-Objective Problem:

Rather than a static heuristic, the acquisition strategy sweeps a Pareto front to simultaneously:

1) **Maximize Explanatory Exploitation:** The MFK provides both a novelty prediction and its uncertainty, allowing us to compute EI_{MFK} (the potential for discovering a highly novel explanation).
2) **Maximize Physical Exploration:** Predictive variance reduction input space $\mathrm{IMSE_{GP}}$.

Dynamic Scalarization: We linearly sweep a Pareto front, starting from spatial uncertainty ($\alpha_{start} = 0.01$) to more explanatory novelty ($\alpha_{end} = 0.9$)

$$O_{Dynamic}(x) = \alpha\, EI_{MFK}\left(S(x)\right) + (1-\alpha)\, \Delta IMSE_{GPs}(x)$$

Pseudocode - initialization

Input: Expensive black-box simulator f, Feature design space $\mathcal{X}$, Initial budget n_{doe}, Total budget N, Low-fidelity explanation budget n_{xpl_lowfi}

Output: Explored dataset $\mathcal{D} = \{x_{doe}, y_{doe}\}$

`// Step 1: Initial Dataset Generation`

Sample n_{doe} points x_{doe} in $\mathcal{X}$ using Latin Hypercube Sampling (LHS);

Evaluate black-box simulator: $y_{doe} \leftarrow f(x_{doe})$;

Initialize dataset $\mathcal{D} \leftarrow \{x_{doe}, y_{doe}\}$;

Initialize dynamic scalarization weight: $\alpha^{(0)} \leftarrow \alpha_{start}$;

Define step size $\Delta\alpha = \frac{\alpha_{end} - \alpha_{start}}{N - n_{doe} - 1}$;

Initialize iteration counter $t \leftarrow 0$;

Pseudocode – active learning

while $|\mathcal{D}| < N$ **do**

// Step 2: Explanation Inputs and Surrogate Training

Train a noisy Gaussian Process (GP) on the current dataset $\mathcal{D}$;

Sample n_{xpl_lowfi} points x_{xpl_lowfi} in $\mathcal{X}$;

Compute ExactSHAP values S for $(x_{xpl\_lowfi} \cup x_{doe})$ using the GP as a proxy, with x_{doe} as the background reference;

// Step 3: Explanation Novelty

Compute proxy novelty y_{nov_total} on the SHAP values S using Cosine Distance (k-NN);

Extract high-fidelity novelties: $y_{nov\_high} \leftarrow y_{nov\_total}(x_{doe})$;

Extract low-fidelity novelties: $y_{nov\_low} \leftarrow y_{nov\_total}(x_{xpl\_lowfi})$;

Train a Multi-Fidelity Gaussian Process (MFGP) to predict explanation density using high-fidelity data $\{x_{doe}, y_{nov\_high}\}$ and low-fidelity data $\{x_{xpl\_lowfi}, y_{nov\_low}\}$;

// Step 4: Dynamic Hybrid Bayesian Optimization

Generate an LHS initial population $\mathcal{P}_{init}$ of size 50 to ensure optimal spatial coverage;

Solve the optimization problem using a global Derivative-Free Mixed-Integer Differential Evolution initialized with $\mathcal{P}_{init}$;

To prevent magnitude domination, objectives are dynamically normalized relative to their maximums observed in $\mathcal{P}_{init}$:;

$$x_{opt} = \arg\max_{x \in \mathcal{X}} \left[(1 - \alpha^{(t)}) \cdot \frac{\Delta \mathrm{IMSE}_{MC}(x)}{\max(\Delta \mathrm{IMSE})} + \alpha^{(t)} \cdot \frac{\widetilde{\mathrm{EI}}_{MFK}(S(x))}{\max(\widetilde{\mathrm{EI}})} \right];$$

// Step 5: Update

Evaluate the expensive black-box at the new point: $y_{opt} \leftarrow f(x_{opt})$;

Update inputs: $x_{doe} \leftarrow x_{doe} \cup \{x_{opt}\}$;

Update outputs: $y_{doe} \leftarrow y_{doe} \cup \{y_{opt}\}$;

Update dataset: $\mathcal{D} \leftarrow \{x_{doe}, y_{doe}\}$;

Update scalarization weight: $\alpha^{(t+1)} \leftarrow \alpha^{(t)} + \Delta\alpha$;

$t \leftarrow t + 1$;

end

Results & Analysis

Experimental Setup

- **Initial Design of Experiments (DoE):** All methods share an identical starting point consisting of a 30-point Latin Hypercube Sample optimized via Enhanced Stochastic Evolutionary (LHS-ESE).
- **Acquisition Strategies Evaluated:**
  - **IMSE-US ($\boldsymbol{\alpha = 0}$):** Pure physical uncertainty reduction.
  - **SHAP-US ($\boldsymbol{\alpha = 1}$):** Pure explanatory novelty search.
  - **Dynamic-US ($\boldsymbol{\alpha \in [0.01, 0, 9]}$):** Our proposed hybrid method, linearly increasing.
- **Internal Optimizer:** The highly non-convex, derivative-free acquisition function $O_{Dynamic}$ is globally maximized at each step using the **Differential Evolution** algorithm (50 starting, 5 iterations) using 250 GP surrogates.
- **Computational Budget:** 50 Active Learning iterations (from $N = 30$ to $N = 80$ total points).
- **Baseline (LHS):** A pure LHS-ESE baseline evaluated iteratively. Crucially, the LHS $(N = 30, \dots 80)$ are all $i.i.d$, fully re-optimized baseline projection to guarantee maximum space-filling efficiency at every step.

The Baseline: LHS-ESE

Establishing a Space-Filling Oracle: We compare our approach against the gold-standard static DoE: Latin Hypercube Sampling optimized via Enhanced Stochastic Evolutionary (LHS-ESE).

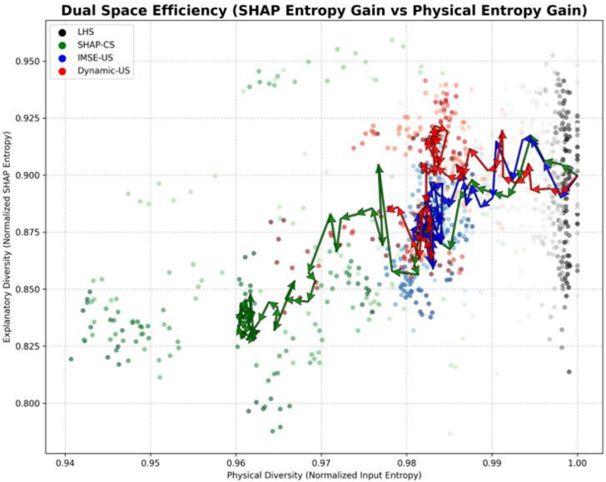
$$\min_{X \in \mathbb{R}^{N \times M}} \left(\sum_{i=1}^{N-1} \sum_{j=i+1}^{N} d\left(x_i, x_j\right)^{-p} \right)^{\frac{1}{p}},$$

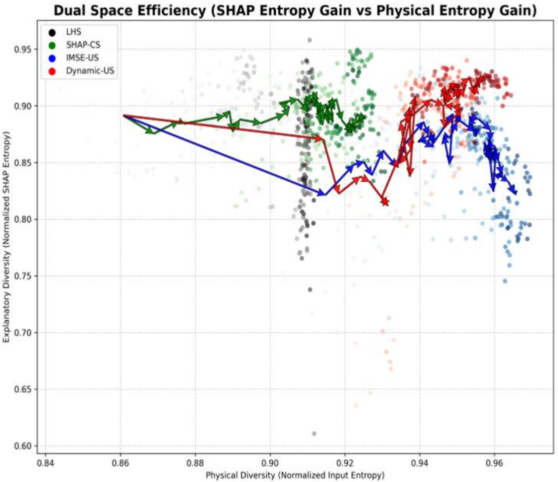
Such that X satisfies the univariate Latin Hypercube constraint.

- **The Solver (Jin et al.):** The discrete combinatorial space of LHS is explored and optimized using the DFO Enhanced Stochastic Evolutionary (ESE) algorithm.
- **Iterative Evaluation:** Our baseline re-solves this constrained problem from scratch at *every step* (from N = 30 to N = 80). This guarantees the absolute mathematical limit of static physical space-filling.

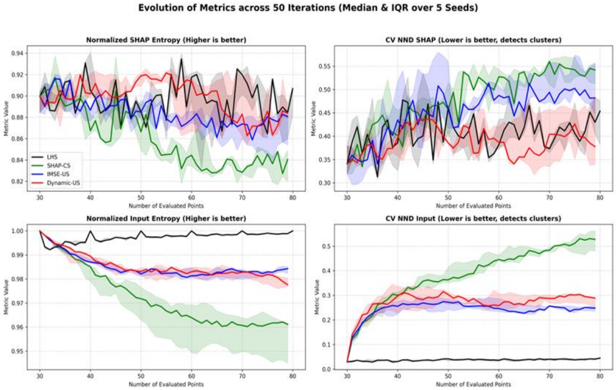
Jin, R., Chen, W., & Sudjianto, A., **An efficient algorithm for constructing optimal design of computer experiments. Journal of statistical planning and inference**, 2005.

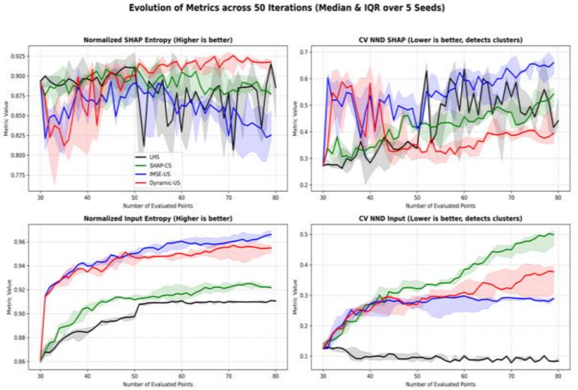
Convergence and Stability over Time



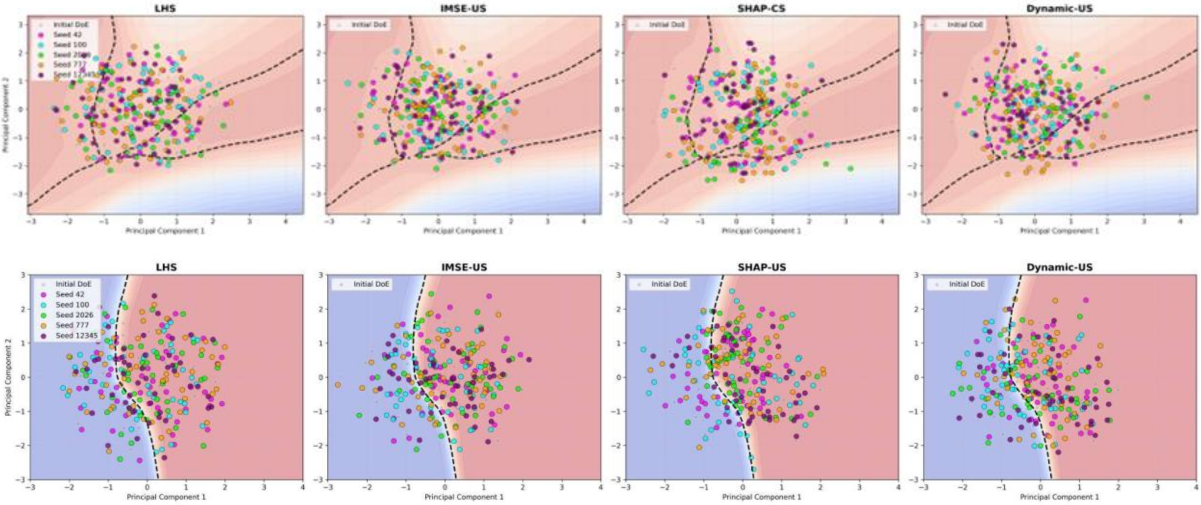


Convergence and Stability over Time

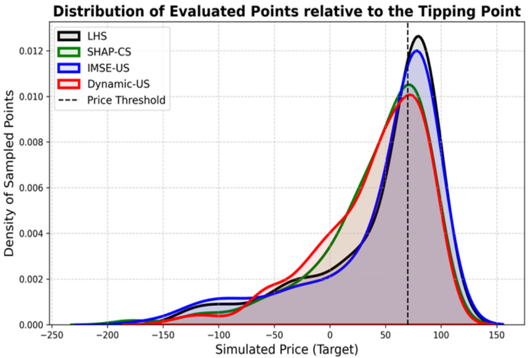


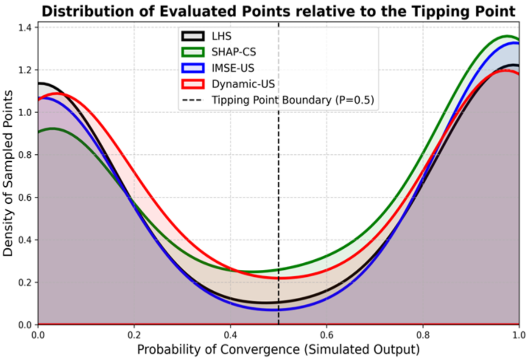


Hunting Tipping Points in PCA Space

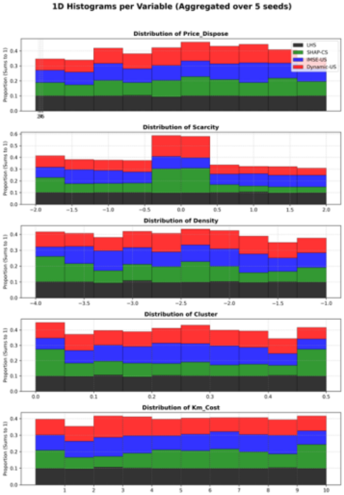


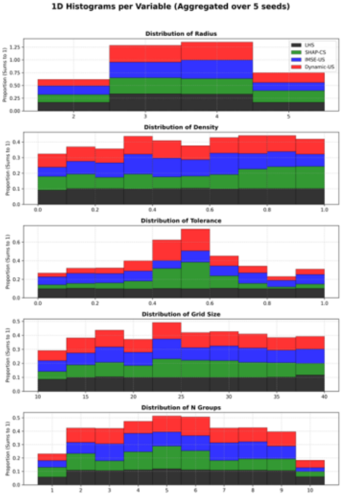
Hunting Tipping Points in the Output Space



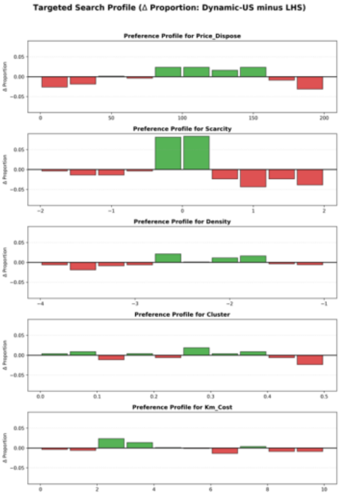


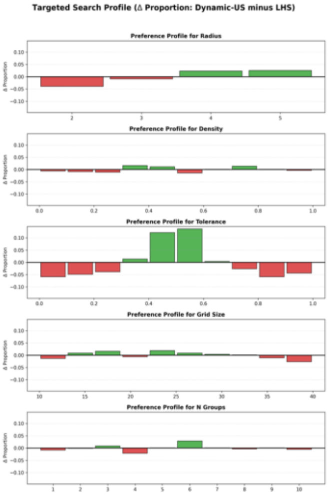
Hunting Tipping Points in the Parameter Space





Hunting Tipping Points in the Parameter Space

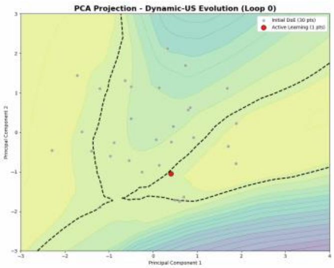


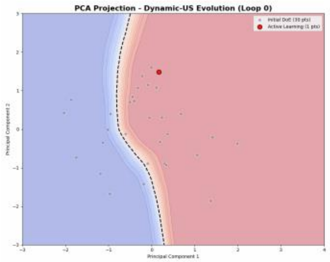


Conclusions & Perspectives

Conclusions

- A model-agnostic, multi-fidelity optimized XAI loop that achieves deep structural understanding rather than just raw predictive accuracy.
- Demonstrates that optimizing for explanatory diversity uncovers tipping points while conserving the classic SUR properties.
- Validated on robust Agent-Based Models (MASTIO, Schelling) to reconcile DFO and XAI.





Perspectives & Future Works

- **Active Subspace Projection:** Linking SHAP tracking to the dynamic rotation of the Active Subspace. This provides a theoretical pathway to scale dimensions by reducing ExactSHAP overhead.
- **Multi-Objective DFO:** Replacing linear scalarization with Expected Hypervolume Improvement to capture non-convex regions of the Pareto front.
- **Trust-Region Bayesian Optimization (TREGO):** Using dynamically sized local trust regions to handle the inherently non-stationary landscapes of SHAP novelty.
- **Global Solvers:** Upgrading the acquisition optimizer from Differential Evolution to CatMADS to avoid local minima traps in the multimodal novelty surface.
- **Optimization under uncertainty:** Put the uncertainty back into the process. Control replicate numbers.
- **Knowledge base:** Add other cases to validate our methodology on more problems. Aggregate metrics across test problems. Generalize to hierarchical problems. Have more repetitions and more iterations.
- **Convergence? Stopping criterion?**
- **Write guidelines to show how DFO can be accounted for in AutoXAI** (going further that static learning)
- **Write the paper – Work on many-objective Multi-fidelity optimization for Canadian transport**

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Thank you for your attention!

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